Is Your Climbing Rope Aging Faster Than You Think?

A summer of sun and rain doesn't make your rope snap, but new research shows it quietly steals away your safety margin, and the coating you paid extra for might be part of the problem.

Reporting on research by D. Sedláček, M. Roso, L. Viel, N. Perotto, B. Caven, M. Hasler, and W. Nachbauer • University of Innsbruck & University of Padova

A climbing rope showing sun-faded sheath colors, a visible sign of UV damage that signals degradation of the internal nylon fibers. Source: Climb On Equipment, "How to Inspect Your Climbing Rope," climbonequipment.com.

Your rope could be quietly failing you, and there's no way to tell just by looking. Most climbers know the routine: you pull your rope out of your bag, run it through your hands checking for soft spots, give it a once-over for fraying, and tie in If it looks fine, it probably is fine, right? According to a team of researchers in Austria and Italy, that assumption may be more dangerous than it seems. The threat isn't a sudden snap or a visible fray, it's a slow, invisible chemical breakdown that no visual inspection can catch.

Climbing ropes have come a long way from their origins. Early mountaineers trusted their lives to twisted hemp ropes that were dangerously prone to snapping under the shock of a real fall. That changed in the 1950s when nylon entered climbing equipment. Unlike hemp, nylon stretches, absorbing fall energy gradually rather than stopping the body all at once. Around the same time, manufacturers developed the kernmantle design, a braided outer sheath wrapped around a load-bearing inner core, which remains the global standard today. By the 1960s, the UIAA, the international body that governs climbing safety standards, had established formal testing procedures including the drop test still used by researchers today. For decades, these advances seemed to have solved the safety problem. But one question kept slipping through the cracks. What happens to a rope after months of sitting in the sun?

The Team Behind the Study

That is precisely what Daniel Sedláček set out to answer. Sedláček, a doctorate-holding researcher at the University of Innsbruck in Austria who works across both sport science and material technology, led a team that included polymer engineer Martina Roso from the University of Padova in Italy, textile chemist Barnaby Caven from the University of Innsbruck, and professional equipment testers from Dolomiticert, a certified climbing gear testing laboratory in Longarone, Italy. Together, the team brought a rare combination of chemistry, materials science, and real-world equipment testing to a question the climbing community had largely left unanswered. Their study examined something most climbers never think about: the slow, invisible chemical breakdown happening inside their rope every time it spends a day outside.

The team had set out to determine whether weathered ropes posed a danger to climbers — but what they found was more complicated than a simple yes or no. Buried in the data was a result that surprised even the researchers themselves: the very coating designed to protect the rope was quietly making part of it worse.

Putting Ropes Through a Summer Outdoors

To find answers, the team used four commercial ropes from Mammut, a well-known Swiss outdoor equipment brand. Two with a hydrophobic Teflon coating and two uncoated. One from each pair was hung outdoors at a mountain weather station near Innsbruck at over 7,400 feet through a full alpine summer, exposed to intense UV radiation, rain, and temperature swings. The other stayed indoors as a control. Every two months, sections were cut and tested using the UIAA 101 drop test, which simulates the most severe fall a climber is likely to take and measures both the force transmitted to the climber and how many such falls the rope can survive before breaking. Individual fibers were also strength-tested, and infrared spectroscopy, a technique that identifies molecular structures by measuring how materials absorb infrared light, was used to detect chemical breakdown at the molecular level.

The Coating That Protects — and Destroys

The results were a mix of reassuring and surprising. The good news is that weathering did not increase the force transmitted to a climber's body during a fall, meaning the immediate injury risk stays stable. But the number of falls each rope could survive before failing told a different story. The uncoated rope dropped from eight falls to four, a fifty percent decline, putting it right at the UIAA minimum standard of five. The coated rope declined from eleven to nine, about twenty percent.

Most counterintuitively, the Teflon coating simultaneously protecting the inner core from water damage while accelerating the breakdown of the outer sheath. When exposed to UV light, the coating releases reactive molecules that attack surrounding nylon fibers and trigger a chain reaction of degradation. The two ropes also changed physically in opposite directions. The coated rope shrank while the uncoated stretched. A difference tied to shifts in the nylon's internal crystal structure that alter how the sheath grips the core during a fall.

What This Means for Climbers

Four months of weather exposure does not make a rope acutely dangerous, but the safety margin shrinks invisibly. No amount of running the rope through your hands will reveal that its fibers have lost significant strength, or that its sheath is degrading from the inside out. The researchers stopped short of prescribing a specific retirement timeline, but their findings make a strong case that climbers should be tracking environmental exposure, not just visible wear, when deciding when to replace a rope.

Future research should examine longer weathering periods and safer coating alternatives, with particular urgency for slings, personal anchors, and accessory cords, which share the same nylon chemistry but lack a protective core entirely, making them potentially even more vulnerable to the degradation this study uncovered. Until then, the safest approach may be to treat your rope like sunscreen. Assume the invisible damage is accumulating, and replace it before you can see the problem.

By the Numbers: 4 Months Outdoors
Falls survived (uncoated rope): 8 → 4 (−50%)
Falls survived (coated rope): 11 → 9 (−20%)
UIAA minimum required falls: 5
Max. impact force change: No significant increase
Test site elevation: ~7,400 ft (2,269 m), near Innsbruck, Austria
Length change: Coated rope shrank ~24 mm/m; uncoated stretched ~22 mm/m

Source: Sedláček et al., "Ageing of climbing ropes with and without hydrophobic coating," Proceedings of the Institution of Mechanical Engineers, Part P.